MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FOURTH SEMESTER MGU-UGP (HONOURS)
REGULAR EXAMINATION, MARCH 2026
MAJOR ELECTIVE – MG4DSEBTS201
RESEARCH METHODOLOGY AND BIOSTATISTICS
(2024 Admission Onwards)

Time: 2hrs **MaximumMarks: 70**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PartA

MultipleChoiceQuestions

Answer **all** Questions
Each question carries **1** mark

1. A unique 13-digit code that identifies a specific book edition is called **[Apply][5]**

a) IBN b)ISSN c)ISBN d) ICBS

2. Research is best defined as: **[Understand][1]**

a) Gathering information from booksb)A systematic investigation to establish facts
c) Collecting data for general use d)Searching the internet for information

3. An arrangement of data in rows and columns is known as ----------. **[Understand][2]**
a) Data collection b) Classification c) Tabulation d) Diagram

4. Which organization publishes the Journal Impact Factor? **[Apply][5]**
a)Scopus b) Google Scholar c)Elsevier d)Clarivate Analytics

5. The h-index measures: **[Apply] [5]**

a)Total number of publications b)Average citations per article

c) Research productivity and citation impact d)Journal ranking

6. Which of the following is NOT a component of a research report? **[Understand][4]**

a) Work plan b)Review of literature c) Data analysis results d) Research methodology

7. Which one of the following is a graph? **[Remember][3]**
a) Bar diagram b) Pictogram c)Histogram d) Pie diagram

8. Test used for comparing the means of two normal populations. **[Understand][3]**

a) F test b)Chi square test c) ANOVA d)Student's t test

9. Data collected from each and every unit of the population is referred to as **[Understand][2]**
a) Primary data b) Census c) Sample d) Secondary data

10. Citation index helps researchers to: **[Apply][5]**

a) Write research proposals b) Track how often a work is cited
c) Edit manuscripts d) Submit to journals

[1 x 10 = 10]

PartB

ShortQuestions

Answer **10** Questions

Eachquestioncarries**3marks**

11. Describe the types of research. **[Remember] [1]**
12. Give a brief account of any two presentation techniques.**[Understand][5]**
13. Give a short note on research reports.**[Understand][4]**
14. What are the advantages of sampling? **[Understand][2, 3]**
15. Differentiatebetweenprimaryandsecondary data**.[Understand][2,3]**
16. Write a short note on the significance of references and citations in a research article. **[Apply][5]**
17. What is a hypothesis? Give a short account.**[Understand][1,2]**
18. Give a short account on different characteristics of research.**[Remember][1]**
19. How to draw a table? **[Remember][2,3]**
20. Define variables. **[Understand][3]**
21. Differentiate bar diagram and Histogram.**[Analyse] [3]**
22. Explain Normal distribution.**[Analyse] [3]**

[10 x 3 = 30]

PartC

Short Essay

Answer **6**Questions

Eachquestioncarries**5marks**

23. Describethe structure and components of a research report.**[Understand][4]**
24. What is a project proposal? Explain the steps involved in preparing a project proposal.**[Apply] [5]**
25. Write an account on scientific communications.**[Understand][5]**
26. What is research? Write a short essay.**[Understand][1]**
27. Explain the types of Classificationof data. **[Understand][2,3]**
28. Explain diagrams and graphs. **[Understand][3]**
29. Explain the Students t test. **[Understand][3]**
30. Explain the chi-square test. **[Analyze][3]**

[6 x 5= 30]